

Auto-ABM: Related Work (Draft)

Anonymous ACL submission

1 Related Work

1.1 The Agent-Based Modeling Ecosystem

Agent-based modeling (ABM) represents a system as autonomous agents that follow local rules and interact with one another and their environment. The modeler specifies individual behavior and observes how macro patterns emerge from the bottom up (Bonabeau, 2002; Macal and North, 2010). Classic studies such as Schelling’s segregation model (Schelling, 1971) and Epstein and Axtell’s Sugarscape (Epstein and Axtell, 1996) established ABM as a standard tool across ecology, epidemiology, economics, and the social sciences. A mature software ecosystem supports this work: NetLogo (Wilensky, 1999), Repast Symphony (North et al., 2013), MASON (Luke et al., 2005), Mesa (Kazil et al., 2020; ter Hoeven et al., 2025), and GAMA (Taillandier et al., 2019). The ODD protocol provides a common standard for documenting models (Grimm et al., 2006, 2020).

Building and studying a model nevertheless remains largely manual work that requires both programming skill and domain knowledge. Researchers still move by hand through reading literature, writing an ODD, translating it into code, debugging, sweeping parameters, and interpreting output (Macal and North, 2010; Grimm et al., 2020). Recent work has brought large language models (LLMs) into isolated steps of this cycle. NetLogo Chat assists with writing and learning NetLogo code (Chen et al., 2024); few-shot prompting and retrieval improve generated code accuracy (Martínez et al., 2024); and SAGE uses a verifier to iteratively produce runnable models (Niu et al., 2024). Connectors built on the Model Context Protocol (Anthropic, 2024), such as NetLogo-MCP (Abbas, 2025), let an assistant create and run models in conversation. Each tool automates a single chore rather than the wider loop in which the simulation, its execution trace, and its ODD are the

objects of study.

1.2 AI-Driven Automation of the Modeling Workflow

A separate line of research asks how far LLM agents can automate multi-step scientific work. ReAct interleaves reasoning with tool use in an iterative loop (Yao et al., 2023); wrapping an LLM with tools, memory, and control logic in an agentic harness yields systems that can plan and execute extended workflows (Wang et al., 2024). This pattern now drives broader AutoResearch efforts. The AI Scientist automates ideation, coding, experimentation, and writing for machine-learning research (Lu et al., 2024); Coscientist plans and runs real chemistry experiments through laboratory automation (Boiko et al., 2023); and AgentEconomist translates economic intuitions into executable experiments over a fixed simulator, grounding each stage in the literature (Chen et al., 2026). These systems span a full study, but they target program code, wet-lab reactions, or one domain-specific simulator rather than the simulation, trace, and ODD that define an ABM research workflow.

Closer to our setting, several systems automate parts of the modeling pipeline from natural language. SAGE and SOCIA generate and repair simulator code from scenario descriptions (Niu et al., 2024; Hua et al., 2025); AgentEconomist orchestrates hypothesis formation, experiment design, and execution over a simulator (Chen et al., 2026). Auto-ABM pursues a broader goal: rather than producing one-off code or automating a single stage, it orchestrates the full research loop around a classical, deterministic ABM. To our knowledge, no prior system combines natural-language workflow orchestration, deterministic execution on a reusable kernel, ODD-aligned documentation, trace-grounded explanation, and one-click reproducible export in a single simulation-first workflow.

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